

Osmania University

REGULAR EXAMINATION ? 2026

Microprocessors and Microcontrollers ? Regular Examination, 2026

Subject Code: EC402 Semester: 4 Regulation: R18

Time: 3 Hours Max. Marks: 70

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Note: This is a demo question paper generated for Backlog Buddy.

Answer all questions. Assume suitable data wherever necessary.
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SECTION A (Unit 1: 8086 Architecture)

1. How is 8086 Architecture implemented? Give an algorithm / procedure and explain. [12 Marks]
2. Derive the expression related to 8086 Architecture and explain each term. [12 Marks]
3. Derive the expression related to 8086 Architecture and explain each term. [12 Marks]

SECTION B (Unit 2: Instruction Set and Assembly Programming)

4. Write short notes on Instruction Set and Assembly Programming. [8 Marks]
5. Solve a problem related to Instruction Set and Assembly Programming. Show all the steps clearly. [12 Marks]
6. Discuss the engineering applications of Instruction Set and Assembly Programming. [12 Marks]

SECTION C (Unit 3: Interrupts and Timers)

7. Describe the working principle of Interrupts and Timers with necessary equations. [12 Marks]
8. Write short notes on Interrupts and Timers. [12 Marks]
9. Derive the expression related to Interrupts and Timers and explain each term. [12 Marks]

SECTION D (Unit 4: Peripheral Interfacing)

10. Solve a problem related to Peripheral Interfacing. Show all the steps clearly. [12 Marks]
11. Compare and contrast the key concepts involved in Peripheral Interfacing. [12 Marks]
12. Describe the working principle of Peripheral Interfacing with necessary equations. [12 Marks]

SECTION E (Unit 5: Microcontroller Architecture)

13. Solve a problem related to Microcontroller Architecture. Show all the steps clearly. [8 Marks]

14. Compare and contrast the key concepts involved in Microcontroller Architecture. [8 Marks]

15. Describe the working principle of Microcontroller Architecture with necessary equations. [12 Marks]

*** End of Question Paper ***